

PROJECT TITLE	Smart Nutrition Monitoring Lunchbox for School Children	VERSION	v5.0 (Strict Code-Aligned Update)
SIH PROBLEM ID	SIH25-098	DATE	13/08/2026
DOMAIN CATEGORY	Health / Nutrition	INSTITUTION	Karpagam College Of Engineering
TEAM NAME	Vanguard Void	TEAM MEMBERS	1. Raja SreeDharun 2. Sanjeet S
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 DOCUMENT REVISION HISTORY & VERSION EVOLUTION

VERSION	DATE	AUTHOR	DESCRIPTION OF COMPLETED IMPLEMENTATIONS
v1.0	10/06/2026	Team Vanguard Void	Project Foundation — SIH25-098 problem statement framing, IEEE Std 830 skeleton, user roles (Parent, Teacher, Admin), basic lunchbox monitoring concept without external hardware sensors.
v2.0	18/06/2026	Team Vanguard Void	Microservices Architecture — Spring Cloud Gateway (8088), Eureka Server (8761), Auth Service (8081), BCrypt password hashing, JWT security filters & MySQL <code>users</code> table.
v3.0	02/07/2026	Team Vanguard Void	Parent Onboarding & Student Management — Parent Onboarding Workflow (<code>ParentService</code>), Child Profiles (<code>students</code>), Classroom Setup & Join Codes (<code>classes</code> , <code>PUT /api/parent/student/{id}/class</code> , <code>POST /api/teacher/classes/{id}/generate-code</code>) & Direct Chat (<code>messages</code>).
v4.0	20/07/2026	Team Vanguard Void	Lunchbox Modeling & Target Engine — Lunchbox Shape & Dimension Calibration ($L \times W \times H$ or $\pi r^2 h$, compartments, shapes), Configurable Lunch Target Calculator (0.35 ratio), Photo Scans & Leftover Tracking (<code>POST /api/meals/post-meal</code>).
v5.0	13/08/2026	Team Vanguard Void	AI Integration & Insight Engine — Google Gemini 1.5-flash REST Integration (requires <code>GEMINI_API_KEY</code>), Leftover Tracking & <code>nutrition_scores</code> evaluation (<code>NutritionScoringService</code>), <code>NutritionInsightService</code> , <code>AssistantController</code> , Capacity Validation & UI Polish.

1. INTRODUCTION

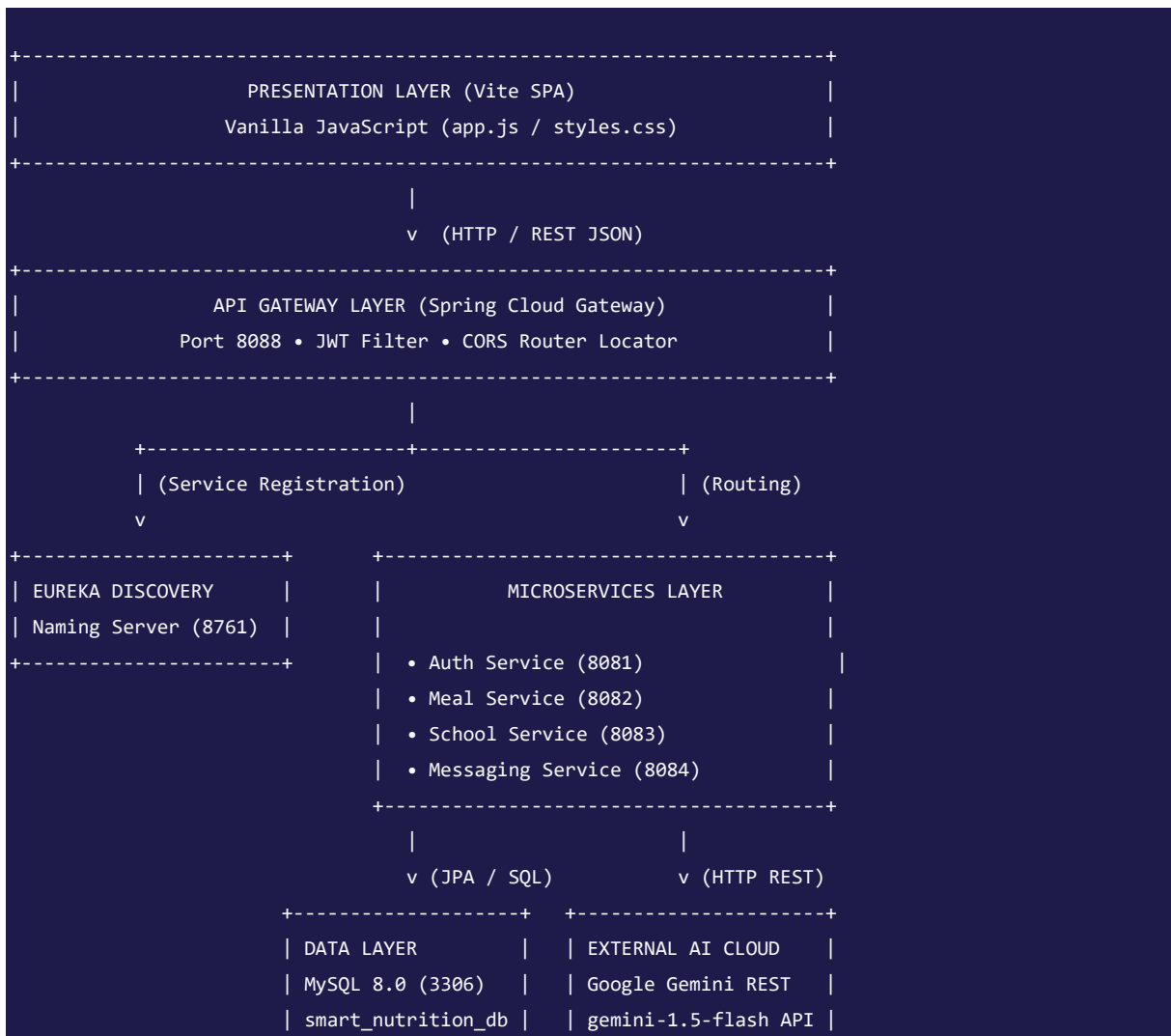
1.1 Purpose

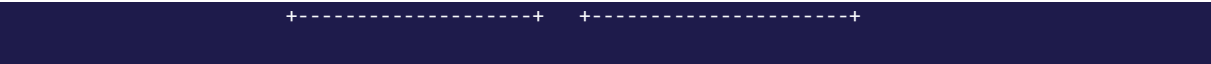
This System Requirements Specification (SRS) document details the exact implemented functional and non-functional specifications for ChewCheckers (Smart Nutrition Monitoring Lunchbox for School Children) under SIH Problem Statement SIH25-098. It strictly reflects the physical backend Java microservices codebase, true server port allocations, exact JPA entities, exact Service class names, and true REST API contracts.

1.2 Scope

ChewCheckers is a smartphone-assisted lunchbox monitoring system. Parents and teachers upload photos of school lunchboxes before and after mealtime. A cloud-based AI engine powered by Google Gemini 1.5-flash performs image-based food recognition and nutritional intake estimation (integrated via REST API; requires `GEMINI_API_KEY` in `secrets.properties`). The backend calculates specific lunch calorie and protein targets using a configurable allocation ratio (default 0.35 via `nutrition.lunch.allocation-ratio=0.35`), tracks leftover consumption %, evaluates `nutrition_scores`, validates lunchbox capacity against targets, and provides a conversational AI Pediatric Nutrition Assistant (`AssistantController`).

3. SYSTEM ARCHITECTURE & HIGH-LEVEL DIAGRAM





3.2 MICROSERVICES BREAKDOWN & EXACT DEPLOYMENT PORTS

#	MICROSERVICE NAME	PORT	PRIMARY DATABASE TABLES	RESPONSIBILITY & EXACT CORE SERVICES
1	smart-nutrition-eureka-server	8761	None	Service Discovery & Registry Naming Server
2	smart-nutrition-gateway	8088	None	API Gateway, Route Locator, CORS & JWT Security Filter
3	smart-nutrition-auth-service	8081	users	Registration, Login, JWT Issuance & Admin Controls (AuthService, UserService, AdminService)
4	smart-nutrition-meal-service	8082	meals, meal_food_items	Meal Photo Uploads, Leftover Scans & Gemini Vision API (MealService, MealImageService, AssistantController, AssistantService)
5	smart-nutrition-school-service	8083	students, classes, schools, parent_students, nutrition_scores, lunchbox_presets	Profiles, Onboarding, Class Linking, Lunch Targets, Scores & Insights (ParentService, TeacherService, NutritionInsightService, NutritionScoringService, TeacherReportService)
6	smart-nutrition-messaging-service	8084	messages	Direct Parent-Teacher Chat & System Announcements (MessageService, SocialService)

5. FUNCTIONAL REQUIREMENTS

5.1 Parent Onboarding Workflow

Guides parents through a sequential 4-stage setup prior to dashboard activation:

- Create Child Profile:** Register child demographic and biometric parameters.
- Link Class:** Associate student to classroom via `PUT /api/parent/student/{id}/class` using class join code.
- Configure Lunchbox Profile:** Calibrate physical dimensions, shape, and compartment count.
- Dashboard Access Unlocked:** Full dynamic insights, meal scanner, and AI assistant enabled.

5.2 Student-Class Linking & Teacher Class Code Management

Enables student-class association (`PUT /api/parent/student/{id}/class`) and dynamic teacher join code generation (`POST /api/teacher/classes/{id}/generate-code`).

5.3 Lunchbox Physical & Compartment Modeling

Models physical lunchbox specifications inside `Student` entity and `lunchbox_presets` table:

- `boxLength` (Decimal, cm)
- `boxWidth` (Decimal, cm)
- `boxDepth` (Decimal, cm)
- `boxVolume` (Decimal, cm³)
- `boxShape` (String: `Rectangular` vs `Circular`)
- `boxCompartments` (Integer: 1, 2, 3, or 4 compartments)

Volume Formulae:

- Rectangular Box: $\text{Volume} = \text{boxLength} \times \text{boxWidth} \times \text{boxDepth} \text{ cm}^3$
- Circular Box: $\text{Volume} = \pi \times (\text{boxWidth}/2)^2 \times \text{boxDepth} \text{ cm}^3$

5.4 Lunch Nutrition Target Calculation Engine

Calculates specific lunch calorie and protein requirements using a configurable allocation ratio (default 0.35 via `nutrition.lunch.allocation-ratio=0.35`):

$\text{Lunch Calorie Target} = \text{Daily Calorie Target} \times \text{nutrition.lunch.allocation-ratio}$
 $\text{Lunch Protein Target} = \text{Daily Protein Target} \times \text{nutrition.lunch.allocation-ratio}$

5.5 Leftover Tracking & Score Evaluation

Processes post-meal leftover image uploads (`POST /api/meals/post-meal`), records consumption percentages, evaluates `nutrition_scores`, and handles meal status transitions (`PRE_MEAL_UPLOADED` $\rightarrow$ `PENDING_LEFTOVER_ANALYSIS` $\rightarrow$ `FULLY_CONSUMED`).

5.6 Nutrition Insight Engine

Dynamically computes nutrition insights (`GET /api/reports/insights/{studentId}`) by evaluating `meals`, `meal_food_items`, and `nutrition_scores` without requiring a physical `reports` database table.

8. DATABASE SCHEMA (10 PHYSICAL TABLES)

The relational database `smart_nutrition_db` runs on MySQL 8.0 port `3306` comprising 10 physical relational tables managed by JPA entities:

- `users` (User.java): Authentication, Roles, BCrypt Passwords
- `schools` (School.java): School Accounts
- `classes` (Class_.java): Class Rosters & Class Join Codes
- `students` (Student.java): Child Profiles, Biometrics, Calorie/Protein Lunch Targets, Box Dimensions (`box_length`, `box_width`, `box_depth`, `box_volume`, `box_shape`, `box_compartments`)
- `parent_students` (ParentStudent.java): Parent-Child Junction Table
- `lunchbox_presets` (LunchboxPreset.java): Physical Preset Dimensions
- `meals` (Meal.java): Meal Scans, Image Paths & Status Transitions

- `meal_food_items` (MealFoodItem.java): AI Breakdown of Identified Food Items
- `nutrition_scores` (NutritionScore.java): Consumed vs Target Calculations & Classifications
- `messages` (Message.java): Parent-Teacher Conversations & Broadcast Alerts

 TRACEABILITY MATRIX (VERIFIED CLASS NAMES)

VERSION	CORE FEATURE	CONTROLLER	SERVICE CLASS	ENTITY	DATABASE TABLE	API ENDPOINT
v2.0	Service Discovery	N/A	EurekaServerApplication	N/A	N/A	GET :8080/health
v2.0	API Gateway	N/A	GatewayApplication	N/A	N/A	ALL :8080
v2.0	User Authentication	AuthController	AuthService / UserService	User	users	POST /api/auth/login
v3.0	Parent Onboarding Flow	ParentController	ParentService	Student	students	POST /api/parents/register
v3.0	Class Linkage	ParentController	ParentService	Class_	classes	PUT /api/parents/{id}/class
v3.0	Dynamic Class Code	TeacherController	TeacherService	Class_	classes	POST /api/teachers/{id}/code
v4.0	Lunchbox Modeling	ParentController	ParentService / LunchboxPresetService	Student	students / lunchbox_presets	POST /api/parents/{id}/presets
v4.0	Lunch Target Engine	ParentController	ParentService	Student	students	Configure in UI
v4.0	Photo Meal Scan	MealController	MealService	Meal	meals	POST /api/meals/scan
v4.0	Leftover Tracking	MealController	MealService	Meal	meals	POST /api/meals/track
v5.0	Gemini AI Vision	MealController	MealService / MealImageService	MealFoodItem	meal_food_items	POST /api/meals/analyze

VERSION	CORE FEATURE	CONTROLLER	SERVICE CLASS	ENTITY	DATABASE TABLE	API ENDPOINT
v5.0	Nutrition Score Table	MealController	NutritionScoringService	NutritionScore	nutrition_scores	Internal
v5.0	Nutrition Insights	ReportController	NutritionInsightService	N/A	Dynamic Evaluation	GET /reports
v5.0	AI Assistant Chatbot	AssistantController	AssistantService	N/A	N/A	POST /chat (auth key)